

Ayesha Najeeb

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EDUCATION

NED University of Engineering & Technology

Oct 2022 – Jun 2026

B.E. Software Engineering | CGPA: 3.8/4.0

- Relevant Coursework: Software Engineering and Design Process, Data Structures and Algorithms, Computer Vision, Data Warehouse & Mining, Networking & Information Security
- Winner, Project Exhibition CodeSphere 2.0 (university-wide software project competition) for the project X-ForNet.

PUBLICATIONS

S. Khan, S. M. Sheraz, M. F. Hyder, Z. Batool, A. Najeeb, "Improving Security in Software-Defined Networks Through an Optimized Support Vector Machine Model for Detecting Distributed Denial-of-Service Attacks," International Journal of Communication Systems, Wiley, Jul 2026. **Link:** <https://doi.org/10.1002/dac.70537>

- Co-developed a Kernel PCA + Genetic Algorithm-optimized SVM for DDoS detection, raising recall from 96.04% to 99.92% and precision from 90.91% to 99.97% on CIC-DDoS2019 dataset.
- Deployed the model in a custom Ryu controller with a FlowTracker for real-time OpenFlow feature extraction and automated flow-rule mitigation, validated on live Mininet traffic

S. Khan, M. F. Hyder, S. M. Sheraz, Z. Batool, A. Najeeb, "X-ForNet: An Explainable AI Framework for Digital Forensics in Software Defined Networking," Concurrency and Computation: Practice and Experience, Wiley (under review)

- Designed a HybridGNN (GraphSAGE, multi-head GAT, GRU) capturing spatial and temporal SDN traffic dependencies, achieving 95.3% detection accuracy.
- Integrated GNNExplainer and GraphXAIN (LLM) to produce human-readable forensic reports that trace attack paths for legal admissibility.

PROFESSIONAL EXPERIENCE

Ontario Tech University | Mitacs Globalink Research Intern

Aug 2026 – Present

- Investigating multi-objective optimization of federated learning for automated cybersecurity in IoT devices and EVCS.
- Benchmarking state-of-the-art federated learning approaches for tree-based models, evaluating predictive performance, computational efficiency, and communication overhead.
- Designed and conducted experiments across CICIDS2017, IoTID2020, and CICEVSE2024 datasets, analyzing experimental results to guide model selection and subsequent research directions.

Folio3 | AI Intern

Oct 2025 – Nov 2025

- Built SmartLearnAI, an agentic AI-based adaptive, personalized learning platform for high school science students, using Generative AI, RAG, and prompt engineering
- Applied LLM-based knowledge graphs to structure course content, improving contextual retrieval for real-time GenAI tutoring and learning evaluation.

Weaves Corporation Limited | Application Development Intern

Jun 2025 – Jul 2025

- Engineered a full-stack Khaadi Rewards App (ASP.NET Core + SQL Server backend, Next.js admin panel, Flutter integration), replacing a third-party vendor solution and cutting operational costs by 40% via AES-encrypted QR transactions.
- Improved product search with Azure OpenAI (GPT-4o, vector embeddings) and semantic ranking, raising retrieval accuracy by 35% with multilingual query support

PROJECTS

BlindAid: Real-Time Blind Assistance System | **Link:** <https://github.com/Ayesha-328/BlindAid> Oct 2025 – Dec 2025

- Designed a real-time assistive pipeline converting live camera input into audio scene descriptions, achieving sub-second scene-to-speech latency.
- Integrated YOLO object detection with a Vision-Language Model to produce contextual natural-language summaries for visually impaired users.

SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks and Development Tools: Agile, Scrum, Figma, Git, GitHub, Vercel, Linux, Azure AI, AWS

Research Areas: Machine Learning, Computer Vision, GenAI, RAG, LLMs, Network Security, SDN, IDS